Experiment 5: Average Fuel Efficiency and Percentage Improvement

Aim

To calculate the average fuel efficiency of different car models and determine the percentage improvement in fuel efficiency between two car models using NumPy arrays and arithmetic operations.

Algorithm

1. Start the program.
2. Import the NumPy library.
3. Create a NumPy array named fuel_efficiency containing the fuel efficiency of different car models.
4. Calculate the average fuel efficiency using np.mean().
5. Calculate the percentage improvement between the first and second car models.
6. Display the average fuel efficiency and percentage improvement.
7. Stop the program.

Input

Car Model 1 = 25 MPG

Car Model 2 = 30 MPG

Car Model 3 = 35 MPG

Car Model 4 = 40 MPG

Car Model 5 = 45 MPG

Code

```
import numpy as np

fuel_efficiency = np.array([25, 30, 35, 40, 45])

average = np.mean(fuel_efficiency)

percentage_improvement = ((fuel_efficiency[1] - fuel_efficiency[0]/ fuel_efficiency[0]) * 100

print("Average fuel efficiency:", average, "MPG")

print("Percentage improvement:", percentage_improvement, "%")
```

Output

```
OneDrive/Documents/FDS/EXP 5.py ====================
=======
Enter fuel efficiency of Car Model 1: 25
Enter fuel efficiency of Car Model 2: 30
Enter fuel efficiency of Car Model 3: 35
Enter fuel efficiency of Car Model 4: 40
Enter fuel efficiency of Car Model 5: 45
Average fuel efficiency: 35.0 MPG
Percentage improvement: 20.0 %
```

Result

Thus, the average fuel efficiency was calculated as 35 MPG, and the percentage improvement between the first and second car models was 20%.